

Solutions to Mathematical Statistics (II) Midterm Exam.

Oct. 22, 2018

1.(50) Let $X_1, \dots, X_{25}$ be r.s. from $N(\mu, 1)$. Consider a test of size $\alpha = 0.05$ for testing $H_0 : \mu = 0$ vs $H_1 : \mu > 0$. Compute power of the test when $\mu = 0.329$ (Hint : $z_{.05} = 1.645$).

(Sol) Under H_0 , $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X}}{1/\sqrt{25}} \sim N(0, 1)$, and we reject if $Z > z_{.05} = 1.645$. Now, the power at $\mu = 0.329$ is $P(\frac{\bar{X}}{0.2} > 1.645) = P(\frac{\bar{X} - 0.329}{0.2} > 1.645 - \frac{0.329}{0.2}) = P(Z > 1.645 - 1.645) = 0.5$.

2.(180) Let $X \sim \text{Exp}(\lambda)$, i.e., the pdf of X is $f(x) = \frac{1}{\lambda}e^{-x/\lambda}I(x > 0)$.

(1)(40) Find the pdf of $T = 1 - e^{-x/\lambda}$.

(Sol) The inverse function is $x = -\lambda \log(1 - t)$, and Jacobian is $\lambda/(1 - t)$. Hence the pdf of T is $g(t) = f(-\lambda \log(1 - t))|J| = I(0 < t < 1)$.

(2)(50) Let $X_1, \dots, X_n$ be r.s. from $\text{Exp}(\lambda)$, and $Y_1 < \dots < Y_n$ be the order statistics of X_i 's. Let $W_1 = nY_1$, $W_2 = (n - 1)(Y_2 - Y_1)$, $W_3 = (n - 2)(Y_3 - Y_2), \dots, W_n = Y_n - Y_{n-1}$. Show that $W_1, W_2, \dots, W_n$ are iid $\text{Exp}(\lambda)$.

(Sol) The inverse functions are $y_1 = \frac{1}{n}w_1$, $y_2 = \frac{1}{n}w_1 + \frac{1}{n-1}w_2, \dots, y_n = \frac{1}{n}w_1 + \frac{1}{n-1}w_2 + \dots + w_n$, and $|J| = 1/n!$. Hence, the jpdf of $W_1, \dots, W_n$ is $g(w_1, \dots, w_n) = f(\frac{1}{n}w_1, \frac{1}{n}w_1 + \frac{1}{n-1}w_2, \dots)/n! = n!\lambda^{-n}e^{-\sum w_i/\lambda}/n! = \prod \frac{1}{\lambda}e^{-w_i/\lambda}$.

(3)(50) Show that $2 \sum_{i=1}^n W_i/\lambda \sim \chi^2(r)$, and find r .

(Sol) Note that $W_i \sim \Gamma(1, \lambda)$, so that $V \equiv \sum W_i \sim \Gamma(n, \lambda)$. Let $Z = 2V/\lambda$, i.e., $v = \lambda z/2$, and $|J| = \lambda/2$. Hence, the pdf of Z is $g(z) = f(\lambda z/2)|J| = \frac{z^{\frac{2n}{2}-1}e^{-z/2}}{\Gamma(\frac{2n}{2})2^{2n/2}} \sim \chi^2(2n)$.

(4)(40) Using the result of (3), find $100(1 - \alpha)\%$ C.I. for λ .

(Sol) Since $2v/\lambda \sim \chi^2(2n)$, we have $1 - \alpha = P(\chi_{1-\frac{\alpha}{2}}^2 \leq \frac{2v}{\lambda} \leq \chi_{\frac{\alpha}{2}}^2)$. Hence, $100(1 - \alpha)\%$ C.I. for λ is $(2v/\chi_{\frac{\alpha}{2}}^2, 2v/\chi_{1-\frac{\alpha}{2}}^2)$.

3.(90) Let $X_1, \dots, X_{10}$ be r.s. from $\text{Poisson}(\theta)$.

(1)(50) Consider a test $H_0 : \theta = 0.2$ vs $H_1 : \theta > 0.2$. Find a randomized test of size $\alpha = 0.05$. (When $Y \sim \text{Poisson}(2)$, $P(Y \leq 4) = 0.947$, $P(Y \leq 5) = 0.983$).

(Sol) Note that $Y = \sum_{i=1}^{10} X_i \sim \text{Poisson}(2)$, and $P(Y \geq 5) = 0.053$, $P(Y \geq$

6) = 0.017. Let $W \sim B(1, p)$ with $P(W = 1) = \frac{.05 - .017}{.053 - .017}$. Then, a randomized test of size $\alpha = 0.05$ is rejecting H_0 if either $\{Y \geq 6\}$ or $\{Y = 5 \text{ \& } W = 1\}$.

(2)(40) Find p -value when $\bar{X} = 0.6$.

(Sol) $P(\bar{X} \geq 0.6) = P(Y \geq 6) = 1 - P(Y \leq 5) = 0.017$

4.(50) Let $X_1, \dots, X_n$ be r.s. from $U(\theta, 1)$, $\theta > 0$. Find the MLE of θ .

(Sol) The likelihood is $L(\theta) = \prod (1 - \theta)^{-1} I(\theta \leq X_i \leq 1) = (1 - \theta)^{-n} I(\theta \leq X_{(1)})$. Now, $(1 - \theta)^{-n}$ is maximized at $X_{(1)}$ when $0 \leq \theta \leq X_{(1)}$.

5.(50) Let $X_1, \dots, X_n$ be r.s. from $U(\theta - 0.5, \theta + 0.5)$. Find the MLE of θ .

(Sol) The likelihood is $L(\theta) = \prod I(\theta - 0.5 \leq X_i \leq \theta + 0.5) = I(\theta - 0.5 \leq X_{(1)} \leq X_{(n)} \leq \theta + 0.5) = I(X_{(n)} - 0.5 \leq \theta \leq X_{(1)} + 0.5)$. Hence, the MLE of θ is any value between $X_{(n)} - 0.5$ and $X_{(1)} + 0.5$.

6.(80) Let $X_1, \dots, X_n$ be r.s. from a distribution with pdf $f(x; \theta)$, $\theta \in R^1$. Let $Y_i = \partial \log f(X_i; \theta) / \partial \theta$, $Z_i = \partial^2 \log f(X_i; \theta) / \partial \theta^2$, $i = 1, \dots, n$.

(1)(40) Show that $\sqrt{n}(\bar{Y} - k_1)$ converges in distribution to $N(0, k_2)$, and find k_1 and k_2 .

(Sol) By the CLT, $k_1 = E(Y_1) = 0$ and $Var(Y_1) = I(\theta)$.

(2)(40) Show that $\bar{Z}$ converges in probability to k_3 , and find k_3 .

(Sol) By the WLLN, $k_2 = E(Z_1) = -I(\theta)$.

Summary Statistics for Midterm Exam. Scores

$n = 63$, $Ave. = 197$

Top Seat : 서유경, 손동현, 임경태, 한정욱 (490 / 500)

